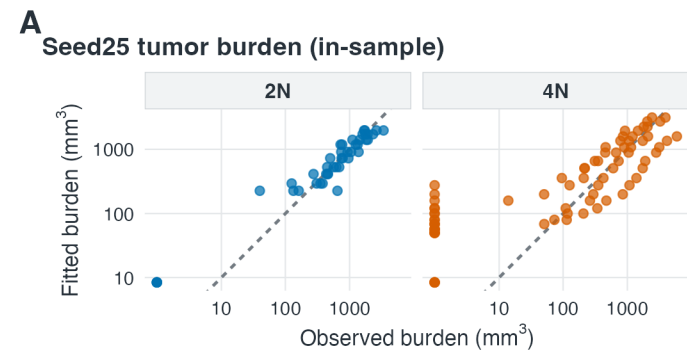
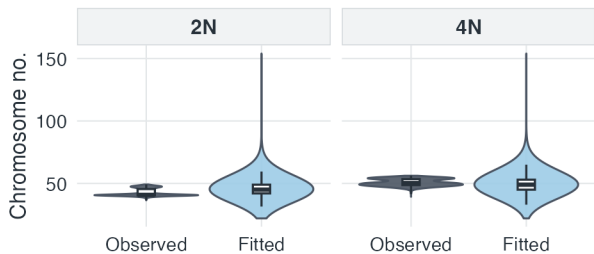


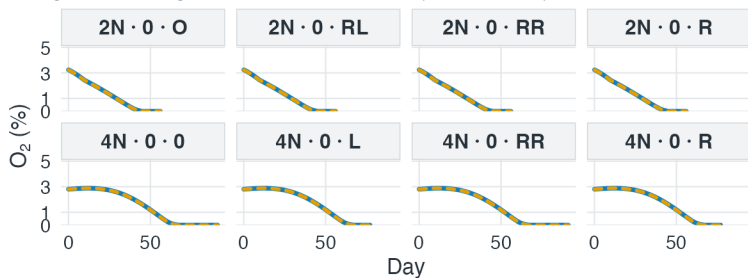


Terminal chromosomes (eight endpoints)



B Target vs latent effective O_2

Orange dashed: target; blue: latent model state (not measured)



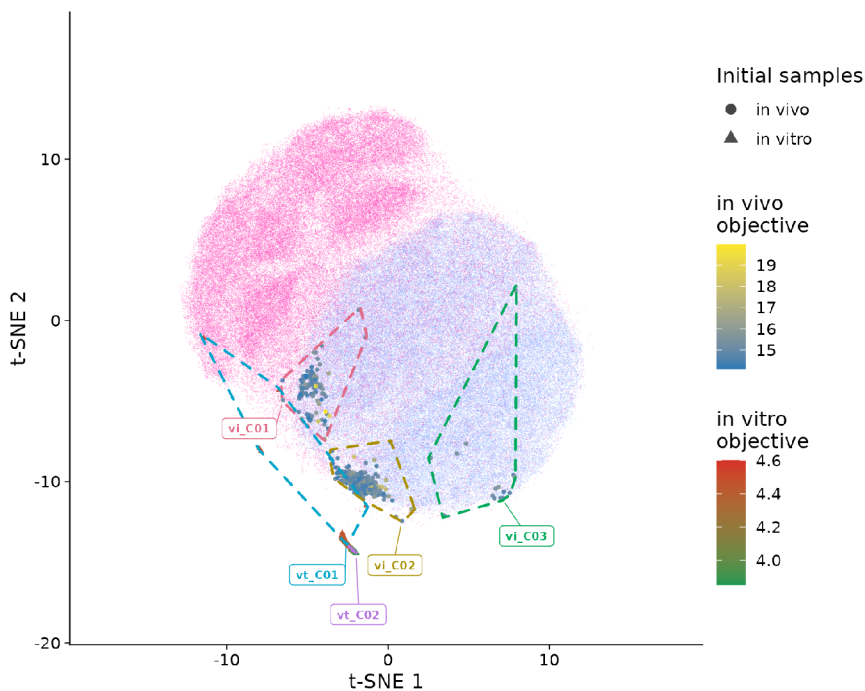
C Univariate fixed- O_2 separators

Discrimination, not causal influence

$\text{O}_2 = 0\%$	$\text{O}_2 = 1\%$	$\text{O}_2 = 5\%$
<i>mu_hp</i>	<i>p_mis_base</i>	<i>p_mis_base</i>
AUC = 0.849	AUC = 0.741	AUC = 0.903
larger: lower-ploidy	larger: higher-ploidy	larger: higher-ploidy

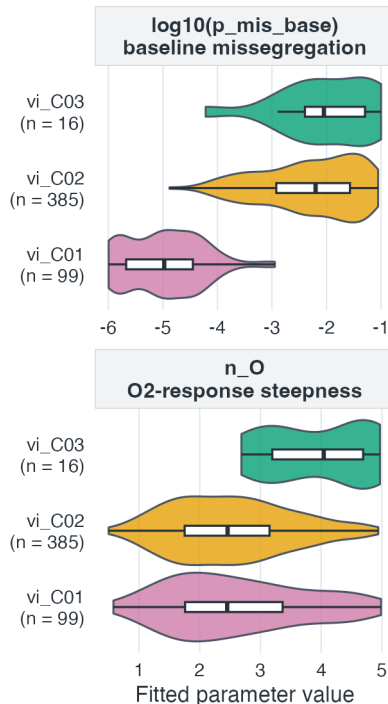
D Pooled parameter embedding

Saved in-vivo/in-vitro coordinates, point universe, and overlays



E Fitted parameters across in-vivo regions

Solution families, not tumor subtypes



A: seed25 is the best weighted-MAP fit among 500 starts; necrosis excluded (no exported predictions).

C: univariate discrimination, not causal influence.

D: approved pooled embedding preserved; whitespace/legend reflow only — no re-embedding or refit.